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Faculty 1 - Institute for Computer Science
Chair for Computer Engineering
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Bachelor/Master Thesis

Thesis title

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Cottbus, 7.4.2026

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Acknowledgments

Abstract

Write your abstract here. This is a concise summary of your thesis, providing readers with an overview of the content, objectives, methodology, results, and conclusions. Avoid including information not found in the thesis itself and keep the abstract to approximately 100–300 words. It must consist of a single paragraph.

Abstract

Verfassen Sie hier Ihren Abstract. Dabei handelt es sich um eine prägnante Zusammenfassung Ihrer Abschlussarbeit, die Ihren Lesern einen Überblick über den Inhalt, die Ziele, die Methodik, die Ergebnisse und die Schlussfolgerungen der Arbeit bietet. Vermeiden Sie es, Informationen zu geben, die in der Arbeit selbst nicht auftauchen und halten Sie sich an eine Länge von ca. 100-300 Wörtern. Der Abstract muss aus einem einzigen Absatz bestehen.

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List of Abbreviations

AI Artificial Intelligence. 1

CPU Central Processing Unit. 1

ML Machine Learning. 1

List of Mathematical Symbols

α_{vc} alpha value for video coding

ϵ epsilon (a small positive quantity)

Chapter 1

Introduction

1.1 Context and motivation

In this thesis, we analyze the performance of Central Processing Unit (CPU) architectures. Modern Machine Learning (ML) techniques often leverage Artificial Intelligence (AI) models to improve accuracy.
Once introduced, CPU will appear only in its short form.
Using classical mechanics, the kinetic energy is defined as shown in Equation (1.1).

$$E_k = \frac{1}{2}mv^2 \tag{1.1}$$

1.2 Challenges and research questions

We denote the error tolerance by epsilon (a small positive quantity) (ϵ). In contrast, the parameter alpha value for video coding (α_{vc}) is used for controlling the video coding process.

1.3 Research methodology

1.3.1 Example Algorithm

Below is an example of a simple algorithm:

Algorithm 1: Sum of the first n numbers

Input: Integer n
Output: Sum of integers from 1 to n
$sum \leftarrow 0$;
for $i \leftarrow 1$ **to** n **do**
    $sum \leftarrow sum + i$;
return sum;

1.3.2 Example Code

Here is an example of inline code using .
To compute the sum in Python, you can use `sum(range(1, n+1))`.
For a quick reference on LaTeX math notation, see the cheat sheet: `https://de.wikipedia.org/wiki/Hilfe:TeX`

1.4 Contributions

1.5 Outline and roadmap

Chapter 2

Background and Related Work

Chapter 3

Methodology

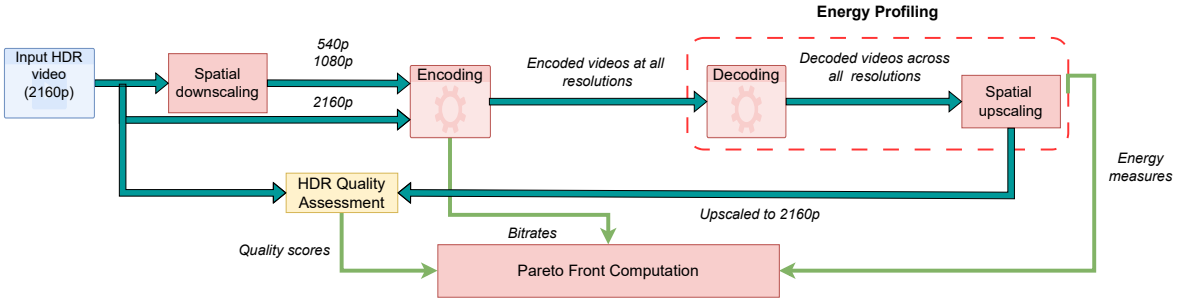


FIGURE 3.1: Overview of the proposed framework.

Chapter 4

Evaluation Setup

TABLE 4.1: Experimental parameters used in this thesis.

Parameter		*Values*					
$\mathcal{R}$	*r [pixels]*	360	432	540	540	540	720
	b [Mbps]	0.145	0.300	0.600	0.900	1.600	2.400
	r [pixels]	720	1080	1080	1440	2160	2160
	b [Mbps]	3.400	4.500	5.800	8.100	11.600	16.800
$\mathcal{C}$		{YUV420, YUV422, YUV444 }					
$\mathcal{D}$		8, 10					
Encoder		*x265 v3.6 [veryslow]*					
Decoder		*HM*					

Chapter 5

Experimental Results

Chapter 6

Conclusions and Future Directions

6.1 Contributions

6.2 Limitations

6.3 Future directions

Appendix A

APPENDIX CHAPTER TITLE

Write your Appendix content here.

Bibliography

[1] F. Bossen, "Common test conditions and software reference configurations," in *JCTVC-L1100*, vol. 12, 2013, p. 7. [Online]. Available: http://phenix.it-sudparis.eu/jct/doc_end_user/current_document.php?id=7281